

# Soohyun Choi

Integrated M.S./Ph.D. Student in Electronic Engineering | Reinforcement Learning  
Hanyang University • Information and Intelligence Systems Lab (IISL) • Seoul, Republic of Korea  
[petersun0221@hanyang.ac.kr](mailto:petersun0221@hanyang.ac.kr) | [schoish.github.io](https://github.com/schoish) | [github.com/SChoish](https://github.com/SChoish)

## Research Interests

My research interests lie in reinforcement learning, particularly in the mathematical and geometric analysis of learning algorithms. I am also interested in developing new algorithms and modeling frameworks using generative models and stochastic processes, especially for goal-conditioned and long-horizon control.

## Education

### Hanyang University

Sep. 2024–Present

*Integrated M.S./Ph.D. Program in Electronic Engineering*

Information and Intelligence Systems Lab (IISL) • Advisor: Prof. Songnam Hong

### Hanyang University

2018–2024

*B.S. in Electronic Engineering • Summa Cum Laude*

## Manuscripts & Research Projects

### PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning

2026

*Soohyun Choi\*, Seonvin Cho\*, and Songnam Hong • Under review • \*Equal contribution • [Code](#)*

Developed a hierarchical offline GCRL framework that connects subgoal selection to short-horizon execution through state-space bridges, inverse dynamics, and execution-time replanning.

### ALARM: Attention and Line Search Based Analog Circuit Optimization on a Riemannian

2024; rev. 2026

#### Manifold

*Soohyun Choi, Minseok Oh, and Ickhyun Song • Undergraduate capstone technical report • [Report](#)*

Formulated designer-selected performance equivalence classes and quotient geometry for circuit search; achieved 10/10 success with 33.2 SPICE evaluations on average in the archived VCO study.

## Ongoing Research

### Multi-step Proximal Policy Improvement

2026–Present

Study re-centered, critic-guided proximal refinement and how step size and refinement depth shape stability across policy-improvement budgets and critic regimes. [\[Code\]](#)

### PathFlower: Goal-Conditioned State Flows

Ongoing

Extend PathBridger toward final-goal-conditioned generative state flows that produce useful intermediate states for long-horizon offline control.

### Adaptive Policy Optimization

Ongoing

Develop adaptive policy refinement that determines when and how far to update under uncertain critic guidance.

## Honors & Funding

### Top Excellence Award - Alumni Association Special Award (KRW 1,000,000)

2023

### Top Excellence Award - 15th Capstone Design Fair (KRW 1,000,000)

2023

### Graduation Honors, Hanyang University

2024

### BK Research Assistantship, Hanyang University

Fall 2025

## Selected Graduate Coursework

**Learning & Control:** Machine Learning; Robot Learning; Advanced Optimal Control Theory

**Probability & Analysis:** Probability and Statistics; Stochastic Processes; Uncertainty Quantification; Functional Analysis I